Weather Forecasting Model Improvement Plan

AI-Driven Weather Forecasting and Crop Recommendation System for Anuradhapura District
22 October 2026 | Proposed plan — no model or production artefact is changed by this document.

1. Purpose and boundaries

This plan improves the existing Python LSTM weather forecasting component in small, reproducible steps. It is based on the project reference set in *docs/source of truth*, *docs/MSc_Project_Documents_PRD_SRS_Architecture*, and *docs/MSc_Project_PRD_SRS_System_Design_Separate_Files*.

The work remains within the documented scope:

- Anuradhapura District representative-point weather data.
- Daily rainfall, mean temperature, and mean relative humidity only.
- A 30-day historical input sequence and direct seven-day forecast.
- An LSTM-based Python model, chronological data splits, MAE/RMSE, and a persistence baseline.
- Clear separation between weather prediction and the crop-suitability engine.

The plan does not add weather variables, agricultural factors, external data providers, evaluation metrics, or a new model family. Such changes require explicit approval.

2. Current baseline

The approved v1 model uses an ERA5-based dataset from 2010-01-01 to 2025-12-31 (5,844 daily observations). Splits are chronological: training 2010–2021, validation 2022–2023, and untouched test period 2024–2025.

Measure	Persistence	LSTM v1
Overall MAE	3.584	3.163
Overall RMSE	8.738	6.729
Rainfall MAE	5.493	4.904
Rainfall RMSE	13.916	10.547
Temperature MAE	0.793	0.707
Humidity MAE	4.465	3.879

The LSTM improves the recorded historical test metrics, but rainfall remains the most difficult output. These are historical ERA5-based model metrics, not live operational accuracy or crop-recommendation accuracy.

3. Improvement roadmap: small to big

Stage 0 — Freeze the baseline and prepare experiment controls

Goal: make every comparison fair and reproducible.

- Preserve v1 model, scaler, metadata, evaluation, and training-history files unchanged.
- Create an experiment register: experiment ID, Git commit, dataset fingerprint, date boundaries, seed, configuration, validation metrics, test metrics, and artefact locations.
- Write every experiment into a new versioned folder; never overwrite v1.
- Add a regression check for split dates, feature order, seven-day horizon, and training-only scaler fitting.

Exit criterion: A fresh baseline run produces comparison-ready metadata and metric reports.

Stage 1 — Diagnose before changing the model

Goal: identify error sources without changing the model.

- Report MAE/RMSE by variable and forecast day for the LSTM and persistence baseline.
- Inspect a small sample of best, typical, and worst historical forecast windows, especially rainfall events.
- Check whether errors concentrate in calendar months or seasonal periods; report only descriptive held-out findings.
- Verify non-negative rainfall after inverse scaling and exact target-date alignment.

Exit criterion: A concise diagnostic report identifies an evidence-based improvement target.

Stage 2 — Low-risk configuration tuning

Goal: improve validation performance while retaining the approved LSTM approach.

Area	Candidate values
Lookback window	14, 30, 60, 90 days
LSTM units	32, 64, 128
LSTM layers	1, 2
Dropout	0.0, 0.1, 0.2, 0.3
Learning rate	0.0005, 0.001, 0.002
Batch size	16, 32, 64

Use the validation period only for selection. Select a candidate only when it improves validation MAE and/or RMSE without material deterioration in the other metric. Document trade-offs.

Stage 3 — Rainfall-preprocessing experiment

Goal: test whether a log1p rainfall transformation reduces the influence of rare extremes while preserving genuine extreme observations.

- Compare current scaling with log1p rainfall transformation followed by scaling.
- Apply the inverse transformation before evaluation and returned inference values.
- Keep temperature and humidity preprocessing unchanged for this isolated test.
- Confirm zero rainfall is handled correctly and no negative rainfall appears after inversion.

Exit criterion: Keep the transform only if validation evidence meets the Stage 2 selection rule.

Stage 4 — Loss-function weighting decision (approval gate)

Goal: consider whether variable contributions to LSTM training should differ. Changing from ordinary mean squared error to a weighted loss is a methodological change; obtain explicit approval before implementation. If approved, compare ordinary and weighted losses only using the winning earlier configuration. MAE and RMSE remain the reported metrics.

Stage 5 — Final candidate confirmation

Goal: perform one honest evaluation of the chosen model on the untouched test period.

- Lock the configuration using validation results only.
- Train once and evaluate on 2024–2025.
- Compare candidate and persistence baseline using overall, per-variable, and per-horizon-day MAE/RMSE.
- Report improvements and regressions; do not retune after seeing the test result.
- Save versioned model, scaler, metadata, evaluation, training history, sample forecasts, and model card.

Stage 6 — Integration, regression testing, and controlled promotion

Goal: safely make a demonstrably better model available to the existing service.

- Confirm compatibility: 30 approved input observations and seven dated output rows.

- Run Python pipeline, inference API, backend integration, and relevant frontend tests.
- Keep v1 as a rollback option.
- Promote only through explicit model-version configuration; never silently replace v1.
- Update README, model evidence, and dissertation material with actual results and limits.

4. Order of execution

- Stage 0: baseline freeze and experiment register.
- Stage 1: diagnostic error analysis.
- Stage 2: small configuration tuning study.
- Stage 3: isolated rainfall-transform study if needed.
- Stage 4: seek approval before weighted-loss testing.
- Stage 5: one final untouched-test evaluation.
- Stage 6: controlled integration and documentation if results justify promotion.

5. Guardrails and stop conditions

- Never use random splitting or fit preprocessing on validation/test data.
- Never report a test metric during configuration selection.
- Never overwrite v1 artefacts.
- Never claim live operational accuracy from historical ERA5 evaluation.
- Stop for approval before changing variables, model family, metrics, provider, or seven-day horizon.
- If no candidate clearly improves validation performance, retain v1 and document the negative result.

6. First implementation package

Begin deliberately small: create experiment configuration/result records; add per-variable and per-horizon diagnostics and tests; rerun the frozen baseline; run only the 14-, 30-, 60-, and 90-day lookback experiments with all other settings equal to v1; then review validation results before expanding the matrix.

7. Source documents consulted

- docs/source of truth/PROJECT_SOURCE_OF_TRUTH_v1.md
- docs/MSc_Project_Documents_PRD_SRS_Architecture/01_Requirements/PROJECT_SOURCE_OF_TRUTH_v1.md
- PRD and SRS working documents in docs/MSc_Project_PRD_SRS_System_Design_Separate_Files/
- docs/phase-5-weather-forecasting.md
- docs/PHASE15_FORECAST_MODEL_EVALUATION.md
- src/forecasting-service/artifacts/v1_era5_2010_2025/v1_evaluation.json

Document limitation: The available working baseline records that the original signed-off proposal was not found in the supplied archive. This plan must be reconciled if that original proposal is later located.